

Partial Differential Equations

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1 Preliminaries

2 Wave Equation in One Dimension

The wave equation in one dimension is

$$u_{tt} = c^2 u_{xx} \tag{1}$$

We first assume the domain is the entire real line. It is easy to see that

$$u_{tt} - c^2 u_{xx} = \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right) u = 0 \tag{2}$$

by completing the square. The solution to the equation on the right is

$$u(x, t) = f(x + ct) + g(x - ct). \tag{3}$$

Thus there exist two functions we can choose for any solution to the wave equation. These solutions correspond to a wavefront moving to the left and right, for the two terms in $u(x, t)$, respectively.

We can also solve the 1D wave equation by a change of coordinates. Let us introduce

$$\xi = x + ct \quad \eta = x - ct \tag{4}$$

After some derivation we get

$$u = f(\xi) + g(\eta) \tag{5}$$

which agrees with our previous solution.

Initial Value Problem

We now introduce the initial value problem (IVP) which takes the form

$$\begin{aligned}u_{tt} &= c^2 u_{xx} \\ u(x, 0) &= \phi(x) \\ u_t(x, 0) &= \psi(x),\end{aligned}\tag{6}$$

with ϕ and ψ arbitrary functions. Let us solve this using (3). First, from $t = 0$ we get

$$\phi(x) = f(x) + g(x)\tag{7}$$

Next we differentiate (3) with respect to t and set $t = 0$ to get

$$\psi(x) = cf'(x) - cg'(x).\tag{8}$$

We first switch the variable we use from x to s for reasons that will become clear in a moment. Then we get from (7) and (8) that

$$\phi' = f' + g' \qquad \frac{1}{c}\psi = f' - g'\tag{9}$$

We can add and subtract these two equations to get

$$f' = \frac{1}{2} \left(\phi' + \frac{\psi}{c} \right) \quad \text{and} \quad g' = \frac{1}{2} \left(\phi' - \frac{\psi}{c} \right).\tag{10}$$

Integrating, we get

$$\begin{aligned}f(s) &= \frac{1}{2}\phi(s) + \frac{1}{2c} \int_0^s \psi + A \\ g(s) &= \frac{1}{2}\phi(s) - \frac{1}{2c} \int_0^s \psi + B,\end{aligned}\tag{11}$$

where A and B are constants. From (7), $A + B = 0$. Thus our final solution, plugging in $s = x + ct$ and $s = x - ct$ for f and g , respectively, is

$$u(x, t) = \frac{1}{2} (\phi(x + ct) - \phi(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s) ds.\tag{12}$$

This is the d'Alembert solution for the one-dimensional wave equation.

Example 2.1. Consider initial conditions that are

$$\phi(x) = \sin(x) \quad \text{and} \quad \psi(x) = 0 \quad (13)$$

Then the second term in (12) disappears and we are left with

$$u(x, t) = \frac{1}{2} (\sin(x + ct) - \sin(x - ct)) = \cos(x) \sin(ct) \quad (14)$$

as our solution. This is exactly a standing wave with speed c .

Example 2.2. Now let us consider the initial conditions

$$\phi(x) = 0 \quad \text{and} \quad \psi(x) = \cos(x) \quad (15)$$

Then the first term in (12) disappears and we get

$$u(x, t) = \frac{1}{2c} (\sin(x + ct) - \sin(x - ct)) = \frac{1}{c} \cos(x) \sin(ct). \quad (16)$$

Notice this is exactly the example above with a $1/c$ term.

Notice that in the first example, the value at a specific x and t depends on the values of ϕ at $x + ct$ and $x - ct$ and only those values. In the second example, instead of just depending on the “edge” values of our initial condition, our solution depends on “all” values by way of our integral. We can demonstrate this difference as follows: consider first an initial condition where ϕ is a large spike centered at $x = 0$ and zero everywhere else (this is the Dirac function). From the d’Alembert solution (12) we know that, for t not equal to zero, the value of u at $x = 0$ must be zero (since $\phi(x + ct)$ and $\phi(x - ct)$ are both 0). Now let us consider an initial disturbance where ψ is the Dirac function. Now, for t not equal to zero, the value of u at $x = 0$ is not zero (in fact, it is 1 , since the integral of the Dirac function for any interval around $x = 0$ is 1). Essentially, it is possible for an initial ψ disturbance to leave a “wake” behind which does not go away (i.e. go to zero and stay there) but is not possible with a ϕ disturbance. It turns out that this idea is very important and the general idea is known as Huygens’ principle.

3 Diffusion Equation in One Dimension

The diffusion equation is

$$u_t = k u_{xx} \quad (17)$$

The Maximum Principle

Consider the diffusion equation in one dimension on a rectangle with $0 \leq x \leq l, 0 \leq t \leq T$. Then the maximum principle says that the diffusion equation takes on its maximum value on this rectangle either at $t = 0$ or on the sides, $x = 0$ or $x = l$.

Proof. We know that at any maximum point on the rectangle it must be that $u_t = 0$, $u_x = 0$, and $u_{xx} \leq 0$. If we knew that $u_{xx} \neq 0$, then the result would follow, since $u_t \neq ku_{xx}$ at that maximum point. We cannot make this statement, though, because u_{xx} can equal zero. Instead we resort to a workaround.

First let M be the maximum value of $u(x, t)$ on the sides $t = 0, x = 0$, and $x = l$. Now let $v(x, t) = u(x, t) + \epsilon x^2$. We try and prove that $v(x, t) \leq M + \epsilon l^2$ for all of R . Then it would follow that $u(x, t) \leq M + \epsilon(l^2 - x^2)$. Let ϵ go to zero and the result follows.

First note that $v(x, t) \leq M + \epsilon l^2$ on $t = 0, x = 0, x = l$. Now, plugging v into the diffusion equation,

$$v_t - kv_{xx} = u_t = k(u + \epsilon x^2)_{xx} = u_t - u_{xx} - 2\epsilon k = -2\epsilon k < 0 \quad (18)$$

Now suppose $v(x, t)$ attains its maximum at a point inside the rectangle, say (x_0, t_0) . Then $v_t = 0$ and $v_{xx} \leq 0$. But then $v_t - kv_{xx} \geq 0$, and we get a contradiction. Now suppose that $v(x, t)$ attains its maximum on the top edge (i.e. $t_0 = T$). We still have $(v_x(x_0, t_0) = 0$ and $v_{xx}(x_0, t_0) \leq 0$. But notice that $v(x_0, t_0) \geq v(x_0, t_0 - \delta)$ for all δ , so we must have $v_t(x_0, t_0) \geq 0$. Thus we still get that $v_t - kv_{xx} \geq 0$, and the proof is complete. $\square$

We can use the same argument with $-u(x, t)$ to get an equivalent statement about the minimum.

The crucial part of this proof is the “diffusion condition” (18) that $v_t - kv_{xx} < 0$, since we can use this condition to derive a contradiction.

The maximum principle can give us an idea of how a solution to the diffusion equation looks like. If we start with any initial condition, the maximum tends to drop down while the minimum comes up. The solution “smooths out” any bumps as time goes on.

Uniqueness

We can use the maximum principle to prove that a solution to the diffusion equation on the rectangle, given initial and boundary conditions, is unique. In other words, any solution to the system

$$\begin{aligned}u_t - ku_{xx} &= f(x, t) & \text{for } 0 < x < l \text{ and } t > 0 \\u(x, 0) &= \phi(x) \\u(0, t) &= g(t) & u(l, t) = h(t)\end{aligned}\tag{19}$$

is unique.

Proof. Let u_1 and u_2 be two solutions to (19). Now let $w = u_1 - u_2$. Then $w_t - kw_{xx} = 0$, $w(x, 0) = 0$, $w(0, t) = 0$, $w(l, t) = 0$. But by the maximum principle (and its equivalent) it must be that $w(x, t) = 0$ for the entire rectangle. But then $u_1 = u_2$, and we are done. $\square$

Solution on the Whole Line

Here we are searching for a solution $u(x, t)$ to

$$\begin{aligned}u_t &= ku_{xx} & \text{with } -\infty < x < \infty, 0 < t < \infty \\u(x, 0) &= \phi(x).\end{aligned}\tag{20}$$

with $\phi(x)$ any function.

We first note a few facts about solutions to (17):

1. If $u(x, t)$ is a solution, then so is the translation $u(x - y, t)$ for any y .
2. Any derivative (or both t and x) of a solution is again a solution.
3. A linear combination of solutions is again a solution.
4. An integral of solutions is again a solution. This is just a limiting case of linearity. In particular,

$$v(x, t) = \int_{-\infty}^{\infty} S(x - y, t)g(y)dy\tag{21}$$

is a solution if $S(x, t)$ is a solution and $g(y)$ is any function, provided the integral converges.

5. If $u(x, t)$ is a solution, then so is the dilation $u(\sqrt{a}x, at)$, for any $a > 0$.
This can be verified by simple substitution.

We can use these facts to construct a general solution to (17). But first, for reasons that will become clear later, we search for a specific solution $Q(x, t)$ which satisfies the initial conditions

$$\begin{aligned} Q(x, 0) &= 1 \quad \text{for } x > 0 \\ Q(x, 0) &= 0 \quad \text{for } x < 0. \end{aligned} \tag{22}$$

The crucial fact about these initial conditions is that they do not change under dilation. Thus, if we have our solution $Q(x, t)$, a dilation of the form $Q(\sqrt{a}x, at)$ does not change the solution at all (since the initial conditions are the same). It is through this intuition that we search for solutions of the form

$$Q(x, t) = g(p) \quad \text{where } p = \frac{x}{\sqrt{4kt}} \tag{23}$$

where we add the $\sqrt{4k}$ term to simplify later calculations. It is clear that Q defined this way does not change under dilation.

We can now find our solution Q by substitution:

$$\begin{aligned} Q_t &= \frac{dg}{dp} \frac{\partial p}{\partial t} = -\frac{1}{2t} \frac{x}{\sqrt{4kt}} g'(p) \\ Q_x &= \frac{dg}{dp} \frac{\partial p}{\partial x} = \frac{1}{\sqrt{4kt}} g'(p) \\ Q_{xx} &= \frac{dQ_x}{dp} \frac{\partial p}{\partial x} = \frac{1}{4kt} g''(p) \\ 0 &= Q_t - kQ_{xx} = \frac{1}{t} \left(-\frac{1}{2} p g'(p) - \frac{1}{4} g''(p) \right) \\ &\Rightarrow g'' + 2p g' = 0. \end{aligned} \tag{24}$$

We can solve this using the integrating factor $\exp \int 2p dp = \exp(p^2)$ to get

$$Q(x, t) = g(p) = c_1 \int e^{-p^2} dp + c_2. \tag{25}$$

Our next step is to find the explicit formula for Q by solving for the con-

stants. We get

$$\begin{aligned}
\text{For } x > 0, \quad 1 &= \lim_{t \rightarrow 0^+} Q = c_1 \int_0^{+\infty} e^{-p^2} dp + c_2 = c_1 \frac{\sqrt{\pi}}{2} + c_2. \\
\text{For } x < 0, \quad 0 &= \lim_{t \rightarrow 0^+} Q = c_1 \int_0^{+\infty} e^{-p^2} dp + c_2 = -c_1 \frac{\sqrt{\pi}}{2} + c_2. \quad (26) \\
\Rightarrow c_1 &= \frac{1}{\sqrt{\pi}} \quad \text{and} \quad c_2 = \frac{1}{2}.
\end{aligned}$$

Thus our solution is

$$Q(x, t) = \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{x/\sqrt{4kt}} e^{-p^2} dp \quad (27)$$

This is officially our final Q .

Having found Q , our next step is to define a new solution $S = \partial Q / \partial x$. This is a solution by the fact that derivatives of solutions are also solutions. Now for the crucial step: we claim that the general solution for (20) is

$$u(x, t) = \int_{-\infty}^{\infty} S(x - y, t) \phi(y) dy \quad \text{for } t > 0. \quad (28)$$

We do not bother here with verifying the solution, which can be accomplished by an integral by parts. Instead, we note that

$$\begin{aligned}
S &= \frac{\partial Q}{\partial x} = \frac{1}{\sqrt{4\pi kt}} e^{-x^2/4kt} \quad \text{and} \\
u(x, t) &= \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{+\infty} e^{-(x-y)^2/4kt} \phi(y) dy. \quad (29)
\end{aligned}$$

This is the final form of our solution.

$S(x, t)$ is the Green's function (or source function) for the diffusion equation in one dimension.

S can be thought of as the solution for u where the initial data is an infinitely large spike (a Dirac function) at the origin. In this view, our solution (28) is simply a stitching of these source solutions together to account for a specific ϕ . It should also be clear now why S is referred to as the source function.

4 Diffusion on the Half Line

We now solve the diffusion equation on the half-line:

$$\begin{aligned} v_t &= kv_{xx} & 0 < t < \infty, \ 0 < x < \infty \\ v(x, 0) &= \phi(x) \\ v(0, t) &= 0 \end{aligned} \tag{30}$$

This problem can be interpreted as, for example, a very long rod with one end in contact with a reservoir of temperature zero.

We solve this problem by reducing it to one we can already solve: diffusion on the whole line. We do so by introducing a new ϕ which is an odd function: i.e. where $\phi(-x) = -\phi(x)$. We form such an odd extension of ϕ that is defined on the whole line as follows:

$$\phi_{odd}(x) = \begin{cases} \phi(x) & \text{for } x > 0 \\ -\phi(-x) & \text{for } x < 0 \\ 0 & \text{for } x = 0. \end{cases} \tag{31}$$

Let $u(x, t)$ be the solution of the new problem

$$\begin{aligned} u_t &= ku_{xx} & \text{for } -\infty < x < \infty, \ 0 < t < \infty \\ u(x, 0) &= \phi_{odd}(x) \end{aligned} \tag{32}$$

The solution is

$$u(x, t) = \int_{-\infty}^{\infty} S(x - y, t) \phi_{odd}(y) dy. \tag{33}$$

The restriction of $u(x, t)$ to $x > 0$ which we deem $v(x, t)$ is a solution of the original problem on the half-line, and by uniqueness is the only one. Thus we have found our v . To find the explicit formula, we have

$$u(x, t) = \int_0^{\infty} S(x - y, t) \phi(y) dy - \int_{-\infty}^0 S(x - y, t) \phi(-y) dy. \tag{34}$$

We can perform a change in variables from $-y$ to y to get

$$u(x, t) = \int_0^{\infty} (S(x - y, t) - S(x + y, t)) \phi(y) dy \tag{35}$$

and finally

$$v(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_0^{\infty} (e^{-(x-y)^2/4kt} - e^{-(x+y)^2/4kt}) \phi(y) dy. \tag{36}$$

Someone very similar can be done for the Neumann problem, where

$$\begin{aligned} w_t &= kw_{xx} && \text{for } 0 < x < \infty, 0 < t < \infty \\ w(x, 0) &= \phi(x) \\ w_x(0, t) &= 0. \end{aligned} \tag{37}$$

This time we use an even extension, where $\phi_{\text{even}}(-x) = \phi_{\text{even}}(x)$. Thus

$$\phi_{\text{even}}(x) = \begin{cases} \phi(x) & \text{for } x \geq 0 \\ \phi(-x) & \text{for } x \leq 0. \end{cases} \tag{38}$$

The solution to this problem is

$$w(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty (e^{-(x-y)^2/4kt} + e^{-(x+y)^2/4kt}) \phi(y) dy \tag{39}$$

by similar reasoning as before.

5 Wave Equation on the Half Line

We now solve the wave equation on the half-line:

$$\begin{aligned} v_{tt} &= c^2 v_{xx} && \text{for } 0 < x < \infty, -\infty < t < \infty \\ v(x, 0) &= \phi(x), & v_t(x, 0) &= \psi(x) \\ v(0, t) &= 0 \end{aligned} \tag{40}$$

We can once again consider the odd extension to the whole line, this time of both ϕ and ψ . Again by similar reasoning we find that the restriction of the odd solution to the half-line is our desired solution:

$$v(x, t) = \frac{1}{2}(\phi_{\text{odd}}(x + ct) + \phi_{\text{odd}}(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi_{\text{odd}}(y) dy. \tag{41}$$

We can understand this solution as follows: first consider $x > c|t|$. Then all arguments occur as positive values, and thus our solution is simply the d'Alembert solution we are already familiar with. If $x < c|t|$, then our solution takes the form

$$v(x, t) = \frac{1}{2}(\phi(ct + x) - \phi(ct - x)) + \frac{1}{2c} \int_{ct-x}^{ct+x} \psi(y) dy. \tag{42}$$

We notice that the argument $ct - x$ is simply the negative of $x - ct$, which we can think of as a “reflection” of the wave which, because of our odd extension, induces a sign change.

Once again, the Neumann problem can be solved by similar reasoning.

6 Diffusion with a Source

We solve the inhomogeneous diffusion equation on the whole line:

$$\begin{aligned} u_t - ku_{xx} &= f(x, t) & \text{for } -\infty < x < \infty, 0 < t < \infty \\ u(x, 0) &= \phi(x) \end{aligned} \quad (43)$$

The source function $f(x, t)$ can be thought of as a source or sink provided to the distribution at different times.

The solution is

$$\begin{aligned} u(x, t) &= \int_{-\infty}^{\infty} S(x - y, t) \phi(y) dy \\ &+ \int_0^t \int_{-\infty}^{\infty} S(x - y, t - s) f(y, s) dy ds. \end{aligned} \quad (44)$$

This second term in the solution can be understood as follows: $S(x - y, t - s)$ can be thought of as a source originating at position y and time s . But this is exactly $f(y, s)$. Thus the solution can be pieced together by simply integrating out all sources contributing to a single point. We will not prove this statement here, but it is simply a verification of the PDE and the IC.

Source on a Half-Line

As an aside consider the more complex problem:

$$\begin{aligned} v_t - kv_{xx} &= f(x, t) & \text{for } 0 < x < \infty, 0 < t < \infty \\ v(0, t) &= h(t) \\ v(x, 0) &= \phi(x) \end{aligned} \quad (45)$$

We can reduce this problem to the inhomogeneous diffusion equation on the whole line by the following substitution: $V(x, t) = v(x, t) - h(t)$. Then V satisfies

$$\begin{aligned} V_t - kV_{xx} &= f(x, t) - h'(t) \\ V(0, t) &= 0 \\ V(x, 0) &= \phi(x) - h(0). \end{aligned} \quad (46)$$

Now all we have to do is apply the method of reflection to find V . We can then recover v as $v(x, t) = V(x, t) + h(t)$.

The Neumann problem

$$\begin{aligned}w_t - kw_{xx} &= f(x, t) \\w_x(0, t) &= h(t) \\w(x, 0) &= \phi(x)\end{aligned}\tag{47}$$

can also be solved in a similar way with $W(x, t) = w(x, t) - xh(t)$.

7 Wave Equation with a Source

The wave equation on the whole line with a source is

$$\begin{aligned}u_{xx} - c^2 u_{tt} &= f(x, t) \\u(x, 0) &= \phi(x) \\u_t(x, 0) &= \psi(x)\end{aligned}\tag{48}$$

The solution of (48) is

$$u(x, t) = \frac{1}{2}(\phi(x + ct) + \phi(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi + \frac{1}{2c} \iint_{\Delta} f \tag{49}$$

where Δ is the characteristic triangle given by the points (x, t) , $(x - ct, 0)$, and $(x + ct, 0)$.

The intuition for (49) should be clear from the analogous solution for the diffusion equation.

We can abstract the method used to obtain the solutions with sources by abstracting away the operators themselves: the method is called the operator method. It turns out solving for the inhomogeneous solution in this way shows us that if we can solve the homogeneous version of a PDE, we can find a solution to the inhomogeneous one as well. This is known as Duhamel's principle.

Source on a Half-Line

Our goal is to solve

$$\begin{aligned}v_{tt} - c^2 v_{xx} &= f(x, t) \quad \text{for } 0 < x < \infty \\v(x, 0) &= \phi(x) \quad v_t(x, 0) = \psi(x) \\v(0, t) &= h(t)\end{aligned}\tag{50}$$

The solution is

$$v(x, t) = \phi \text{ term} + \psi \text{ term} + \frac{1}{2c} \iint_D f + h(t - \frac{x}{c}) \quad (51)$$

where D is the characteristic region as in the reflected wave. The new term is $h(t - x/c)$. I like to think of this term as like a wave that travels to the right by imagining h as a function on the negative x -axis.

We can derive this h term as follows: let $\phi = \psi = f = 0$. Now let $v(x, t) = j(x + ct) - g(x - ct)$, which we can always do by (2) and by that $f = 0$. Now, from $\phi = \psi = 0$, we get that $j(s) = g(s) = 0$ for $s > 0$. Then $h(t) = v(0, t) = g(-ct)$ (since j vanishes), and thus $g(s) = h(-s/c)$. Finally we get that $v(x, t) = 0 + h(-(x - ct)/c) = h(t - x/c)$.

8 Fourier's Method

So far we have solved PDEs explicitly by finding a formula based on the ICs and BCs. This has led to relatively nice and intuitive formulae. But certain conditions, for example the physically realistic case of a finite interval (such as $0 < x < l$), lead to much more complicated expressions for solutions. This leads us to the Fourier method.

Separation of Variables

We first consider the wave equation under Dirichlet conditions:

$$\begin{aligned} u_{tt} &= c^2 u_{xx} & \text{for } 0 < x < l \\ u(x, 0) &= \phi(x) & u_t(x, 0) = \psi(x) \\ u(0, t) &= 0 = u(l, t) \end{aligned} \quad (52)$$

We consider solutions of the form

$$u(x, t) = X(x)T(t) \quad (53)$$

Obviously not all solutions have to take this form, but we will see that all solutions must be an infinite sum of such separable solutions.

Plugging this solution form into the wave equation, we get

$$\begin{aligned} X(x)T''(t) &= c^2 X''(x)T(t) \\ \Rightarrow -\frac{T''}{c^2 T} &= -\frac{X''}{X} = \lambda \end{aligned} \quad (54)$$

where we divide by $-c^2XT$ and λ is an arbitrary constant. We note that the two expressions must be equal to a constant because they both only depend on either X or T .

We let $\lambda = \beta^2$ where $\beta > 0$. Then we get two equations, one for X and one for T ,

$$\begin{aligned} X'' + \beta^2 X &= 0 \\ T'' + c^2 \beta^2 T &= 0 \end{aligned} \tag{55}$$

These are easily solvable, and the solutions are

$$\begin{aligned} X(x) &= C \cos \beta x + D \sin \beta x \\ T(t) &= A \cos \beta ct + B \sin \beta ct \end{aligned} \tag{56}$$

where A , B , C , and D are constants.

Now we impose our Dirichlet boundary conditions and get that $X(0) = 0 = X(l)$. Thus it must be that $C = 0$ and $0 = X(l) = D \sin \beta l$, and we get $\beta l = n\pi$. That is,

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin \frac{n\pi x}{l} \quad \text{for } n = 1, 2, 3, \dots \tag{57}$$

Each n provides a solution to (52), so by combining our separated solutions, we get a sequence of solutions

$$u_n(x, t) = \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l} \tag{58}$$

where we used that $\beta = n\pi/l$. But since the wave equation is linear, we can also combine any two solutions u_n to get a third, and thus any sum of these separable solutions must also be a solution. Thus the most general form we can achieve is

$$u(x, t) = \sum_n \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l} \tag{59}$$

From here we get to the crucial question, one asked by Fourier in the 19th century, of whether any pair of problem conditions ϕ , ψ , can be formed by such a (potentially infinite) sum. What was eventually found, and proven, was that any initial condition ϕ (and thus also ψ) can be represented as an infinite sum of these sine (or cosine) functions. These are the famous Fourier series which is an entire topic on its own.

We can also repeat this process for the diffusion equation,

$$\begin{aligned} u_t &= k u_{xx} && \text{for } 0 < x < l, 0 < t < \infty \\ u(0, t) &= u(l, t) = 0 \\ u(x, 0) &= \phi(x) \end{aligned} \tag{60}$$

We have $u(x, t) = X(x)T(t)$, and we get

$$\frac{T'}{kT} = \frac{X''}{X} = -\lambda \tag{61}$$

For T , we get $T' = -\lambda k T$ whose solution is $T(t) = A e^{-\lambda k t}$. The problem for X is exactly the same as for the wave equation. Our final solution is

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-(n\pi/l)^2 k t} \sin \frac{n\pi x}{l} \tag{62}$$

(where $\lambda = \beta^2$).

The numbers λ_n are called eigenvalues and the X_n are called eigenfunctions. This is because, if we treat differentiation as an operator analogous to linear algebra, the differential equation turns into an eigenvalue problem. (In fact, since functions can be thought of as an infinite-dimensional vector space, we can say we are here dealing with infinite-dimensional linear algebra).

A similar process can be carried out for Neumann and Robin conditions, with the Robin being slightly more complex.

9 Harmonic Functions

Laplace's Equation

Laplace's equation is

$$\begin{aligned} u_{xx} &= 0 && \text{in one dimension} \\ \nabla^2 u = \Delta u = u_{xx} + u_{yy} &= 0 && \text{in two dimensions} \\ \nabla^2 u = \Delta u = u_{xx} + u_{yy} + u_{zz} &= 0 && \text{in three dimensions} \end{aligned} \tag{63}$$

Poisson's equation is

$$\Delta u = f. \tag{64}$$

In one dimension, it is clear that $u(x) = A + Bx$, i.e. it is a linear equation. This is clear from $u_{xx} = 0$.

Maximum Principle. Let D be a connected bounded open set (in 2D or 3D). Then if u is harmonic on D that is continuous on $\overline{D} = D \cup \text{bdy } D$. Then, unless u is constant, then the maximum and minimum values are achieved on $\text{bdy } D$.

Proof. The proof is very similar to the one for the diffusion equation. We assume we are working in two dimensions. The idea is that if a point is maximum inside D , then $u_{xx} \leq 0$ and $u_{yy} \leq 0$ and thus $u_{xx} + u_{yy} \leq 0$. Once again, if we can show that $u_{xx} < 0$ and $u_{yy} < 0$ then we would be done. But it is possible for both to equal zero, so we need to be clever again.

Let $v(\mathbf{x}) = u(\mathbf{x}) + \epsilon|\mathbf{x}|^2$. Then

$$\Delta v = \Delta u + \epsilon\Delta(x^2 + y^2) = 0 + 4\epsilon > 0 \quad (65)$$

But for an interior maximum point, $v_{xx} + v_{yy} \leq 0$. Thus v has no interior maximum in D .

Let $\mathbf{x}_0$ be the maximum for v on $\text{bdy } D$. Then

$$u(\mathbf{x}) \leq v(\mathbf{x}) \leq v(\mathbf{x}_0) = u(\mathbf{x}_0) + \epsilon|\mathbf{x}_0|^2 \leq \max_{\text{bdy } D} u + \epsilon l^2, \quad (66)$$

where l is the maximum distance from the origin to the border of D . Thus, letting $\epsilon \rightarrow 0$,

$$u(\mathbf{x}) \leq \max_{\text{bdy } D} u \quad (67)$$

and we are done. $\square$

We can once again do the same for the minimum value.

Invariance in Two Dimensions. The Laplace equation is invariant under all rigid motions, where a rigid motion is either a translation or a rotation. By translation we mean $x' = x + a$ and $y' = y + b$, and a rotation is $x' = x \cos \alpha + y \sin \alpha$ and $y' = -x \sin \alpha + y \cos \alpha$. The proof is rather simple, just substitute the transformations into the laplace equation.

Since Laplace's equation is invariant under rotations, we might suspect that using polar coordinates might be advantageous. This is our next task. The coordinate transformation is

$$x = r \cos \theta \quad y = r \sin \theta \quad (68)$$

The Jacobian matrix is

$$J = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{pmatrix} \quad (69)$$

and the inverse is

$$J^{-1} = \begin{pmatrix} \frac{\partial r}{\partial x} & \frac{\partial \theta}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial \theta}{\partial y} \end{pmatrix} = \begin{pmatrix} \cos \theta & -\frac{\sin \theta}{r} \\ \sin \theta & \frac{\cos \theta}{r} \end{pmatrix} \quad (70)$$

Thus we get

$$\begin{aligned} \frac{\partial}{\partial x} &= \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}, \\ \frac{\partial}{\partial y} &= \sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \end{aligned} \quad (71)$$

and

$$\begin{aligned} \frac{\partial^2}{\partial x^2} &= \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right)^2 \\ &= \cos^2 \theta \frac{\partial^2}{\partial r^2} - 2 \left(\frac{\sin \theta \cos \theta}{r} \right) \frac{\partial^2}{\partial r \partial \theta} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2}{\partial \theta^2} \\ &\quad + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial}{\partial \theta} + \frac{\sin^2 \theta}{r} \frac{\partial}{\partial r} \\ \frac{\partial^2}{\partial y^2} &= \left(\sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \right)^2 \\ &= \sin^2 \theta \frac{\partial^2}{\partial r^2} + 2 \left(\frac{\sin \theta \cos \theta}{r} \right) \frac{\partial^2}{\partial r \partial \theta} \\ &\quad + \frac{\cos^2 \theta}{r^2} \frac{\partial^2}{\partial \theta^2} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial}{\partial \theta} + \frac{\cos^2 \theta}{r} \frac{\partial}{\partial r}. \end{aligned} \quad (72)$$

Finally, we get

$$\Delta_2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \quad (73)$$

As an aside, let us find harmonic functions that are themselves rotationally invariant, or in other words depend only on r . Removing the other terms in (73) we get

$$0 = u_{rr} + \frac{1}{r} u_r \quad (74)$$

which is easily solvable:

$$(ru_r)_r = 0, \quad ru_r = c_1, \quad u = c_1 \log r + c_2. \quad (75)$$

Invariance in Three Dimensions. Rotational invariance in three dimensions can also be shown: we will not do so here. It is important to note some important results which we will also not derive here: first the three-dimensional spherical coordinates Laplacian:

$$\Delta_3 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}. \quad (76)$$

We once again look for solutions invariant under rotations, and thus depend only on r . Here we get

$$0 = \Delta_3 u = u_{rr} + \frac{2}{r} u_r. \quad (77)$$

Thus $(r^2 u_r)_r = 0$. The solutions are $r^2 u_r = c_1$, or $u = -c_1 r^{-1} + c_2$.

Special Geometries

Special geometries, such as rectangles and cubes or circles, may produce explicit solutions to the Laplace equation. For example, Laplace's equation on a rectangle can be solved through separation on variables and applying the suitable BCs from the geometry. The calculation is straightforward and will not be reproduced here.

A more interesting case is the Dirichlet problem for a circle, which takes the form (in two dimensions)

$$\begin{aligned} u_{xx} + u_{yy} &= 0 & \text{for } x^2 + y^2 < a^2 \\ u &= h(\theta) & \text{for } x^2 + y^2 = a^2 \end{aligned} \quad (78)$$

Let us first transfer the problem to polar coordinates, then apply separation of variables $u = R(r)\Theta(\theta)$. Then

$$\begin{aligned} 0 &= u_{xx} + u_{yy} = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} \\ &= R''\Theta + \frac{1}{r} R'\Theta + \frac{1}{r^2} R\Theta''. \end{aligned} \quad (79)$$

Multiplying by $r^2/R\Theta$, we get

$$\begin{aligned}\Theta'' + \lambda\Theta &= 0 \\ r^2 R'' + rR' - \lambda R &= 0.\end{aligned}\tag{80}$$

For $\Theta(\theta)$ we naturally require that $\Theta(\theta + 2\pi) = \Theta(\theta)$, and so

$$\lambda = n^2 \quad \Theta(\theta) = A \cos n\theta + B \sin n\theta \quad \text{for } n = 1, 2, \dots \tag{81}$$

The equation for R is of Euler type with $R(r) = r^\alpha$. Since $\lambda = n^2$ we get for R

$$\alpha(\alpha - 1)r^\alpha + \alpha r^\alpha - n^2 r^\alpha = 0 \tag{82}$$

and thus $R(r) = Cr^n + Dr^{-n}$. The full u is

$$u = \left(Cr^n + \frac{D}{r^n} \right) (A \cos n\theta + B \sin n\theta). \tag{83}$$

Now, since n can be zero we need another linearly independent solution, of which we can take $R = \log r$, which are solutions of the form

$$u = C + D \log r. \tag{84}$$

Our next step is to reject solutions we do not want. First we reject solutions that grow as r^{-n} or $\log r$ since they blow up at the origin. Then our solutions become

$$u = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} (A_n \cos n\theta + B_n \sin n\theta) \tag{85}$$

Now we use the BCs at $r = a$. This means that

$$h(\theta) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} a_n (A_n \cos n\theta + B_n \sin n\theta) \tag{86}$$

This is simply a Fourier series for $h(\theta)$, and so we know that

$$\begin{aligned}A_n &= \frac{1}{\pi a^n} \int_0^{2\pi} h(\phi) \cos n\phi d\phi \\ B_n &= \frac{1}{\pi a^n} \int_0^{2\pi} h(\phi) \sin n\phi d\phi.\end{aligned}\tag{87}$$

This is satisfactory enough, but we can do better. It turns out that combining (85) and (87), we can sum the terms explicitly.

$$\begin{aligned}
u(r, \theta) &= \int_0^{2\pi} h(\phi) \frac{d\phi}{2\pi} \\
&\quad + \sum_{n=1}^{\infty} \frac{r^n}{\pi a^n} \int_0^{2\pi} h(\phi) (\cos n\phi \cos n\theta + \sin n\phi \sin n\theta) d\phi \\
&= \int_0^{2\pi} h(\phi) \left[1 + 2 \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n \cos n(\theta - \phi) \right] \frac{d\phi}{2\pi}.
\end{aligned} \tag{88}$$

The term in the brackets can explicitly be summed as follows:

$$\begin{aligned}
&1 + \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n e^{in(\theta - \phi)} + \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n e^{-in(\theta - \phi)} \\
&= 1 + \frac{re^{i(\theta - \phi)}}{a - re^{i(\theta - \phi)}} + \frac{re^{-i(\theta - \phi)}}{a - re^{-i(\theta - \phi)}} \\
&= \frac{a^2 - r^2}{a^2 - 2ar \cos(\theta - \phi) + r^2}
\end{aligned} \tag{89}$$

Thus (85) can be written as

$$u(r, \theta) = (a^2 - r^2) \int_0^{2\pi} \frac{h(\phi)}{a^2 - 2ar \cos(\theta - \phi) + r^2} \frac{d\phi}{2\pi}. \tag{90}$$

In fact, we can put this in a more intuitive form. First let $\mathbf{x}$ be a point anywhere in our domain, and let $\mathbf{x}'$ be a point on the boundary of the circle. Now, using the origin, $\mathbf{x}$, and $\mathbf{x}'$, we can use the law of cosines to get

$$|\mathbf{x} - \mathbf{x}'|^2 = a^2 + r^2 - 2ar \cos(\theta - \phi) \tag{91}$$

and since $ds' = ad\phi$ (where ds' is a small segment on the circumference), we can put (90) as

$$u(\mathbf{x}) = \frac{a^2 - |\mathbf{x}|^2}{2\pi a} \int_{|\mathbf{x}'|=a} \frac{u(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^2} ds' \tag{92}$$

(where $u(\mathbf{x}') = h(\phi)$). This is Poisson's formula.

Mean Value Property. We can use Poisson's formula to prove a number of things. First is the mean value property, which says that the value of u at the center of a disk D is the average of u on the circumference.

Proof. Letting $\mathbf{x} = \mathbf{0}$ in (92) we get

$$u(\mathbf{0}) = \frac{a^2}{2\pi a} \int_{|\mathbf{x}'|=a} \frac{u(\mathbf{x}')}{a^2} ds'. \quad (93)$$

This is simply the average of u on the circumference. $\square$

We can also use the mean value property to prove the maximum principle that we showed before.

10 Green's Identities and Functions

Green's First Identity

We start from

$$(vu_x)_x = v_x u_x + vu_{xx} \quad (94)$$

which we can generalize to three dimensions to

$$\nabla \cdot (v \nabla u) = \nabla v \cdot \nabla u + v \Delta u. \quad (95)$$

We now integrate and apply the divergence theorem to get

$$\iint_{\text{bdy } D} v \frac{\partial u}{\partial n} dS = \iiint_D \nabla v \cdot \nabla u d\mathbf{x} + \iiint_D v \Delta u d\mathbf{x}, \quad (96)$$

where n is the direction normal to the boundary surface. This is Green's first identity.

Mean Value Property We can use Green's first identity to prove the mean value property in three dimensions (we have already done so in two).

Proof. Assume we are working in a sphere with radius a . First notice that

$$\frac{\partial u}{\partial n} = \frac{\partial u}{\partial r} \quad (97)$$

since we are working in a sphere. Now use Green's first identity with $v = 1$ to get

$$\iint_{\text{bdy } D} \frac{\partial u}{\partial r} dS = 0. \quad (98)$$

In spherical coordinates this is

$$\int_0^{2\pi} \int_0^\pi u_r(a, \theta, \phi) a^2 \sin \theta d\theta d\phi = 0. \quad (99)$$

We now divide by $4\pi a^2$ (so that the value is independent of the area of $\text{bdy}D$). Now we can vary a and treat it as a variable in this equation. Let's call it r . Then

$$\frac{\partial}{\partial r} \left[\frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi u(r, \theta, \phi) \sin \theta d\theta d\phi \right] = 0. \quad (100)$$

This means that the expression in the brackets must be independent of r . This expression is exactly the average value of u on the sphere with radius r . But, letting $r \rightarrow 0$, we get that the expression in brackets is exactly $u(\mathbf{0})$. Since this value does not change as we increase r , we are done. $\square$

Again, we can use the mean value property in three dimensions to prove the maximum principle in three dimensions.

Green's Second Identity

We start from Green's first identity and notice that we can exchange u and v . If we do so and subtract the two equations we get

$$\iint_{\text{bdy}D} \left(v \frac{\partial u}{\partial n} - u \frac{\partial v}{\partial n} \right) dS = \iiint_D (v \Delta u - u \Delta v) d\mathbf{x}. \quad (101)$$

This is Green's second identity.

We can then form the following definitions: a boundary condition is called symmetric for the operator Δ if the left side of (101) vanishes for any choice of functions u and v that satisfy the boundary condition. It is easy to verify that the Dirichlet, Neumann, and Robin BCs are all symmetric.

Representation Formula. We can use Green's second identity to construct a formula for any harmonic function on a domain D based on the values at the boundary. The formula is

$$u(\mathbf{x}_0) = \iint_{\text{bdy}D} \left[-u(\mathbf{x}) \frac{\partial}{\partial n} \left(\frac{1}{|\mathbf{x} - \mathbf{x}_0|} \right) + \frac{1}{|\mathbf{x} - \mathbf{x}_0|} \frac{\partial u}{\partial n} \right] \frac{dS}{4\pi} \quad (102)$$

Proof. We first assume $\mathbf{x}_0$ is the origin, which we can always do. We choose a domain of D_ϵ that is exactly D minus a small ball with radius ϵ and $\mathbf{x}_0$ as its center. We choose this because we would like to exclude the point $\mathbf{x}_0$ because it is infinite for the function $v(\mathbf{x}) = (-4\pi|\mathbf{x} - \mathbf{x}_0|)^{-1}$. Taking this as our domain, we may apply Green's second identity. Since $\Delta u = 0$ and $\Delta v = 0$ on D_ϵ , the triple integral side of Green's identity is simply zero, and we are left with the integral on the boundary. We have two boundaries, one the boundary of D and one we introduced by carving out our sphere. Therefore Green's identity reads

$$-\iint_{\text{bdy}D} \left[u \cdot \frac{\partial}{\partial n} \left(\frac{1}{r} \right) - \frac{\partial u}{\partial n} \cdot \frac{1}{r} \right] dS = -\iint_{r=\epsilon} \left[u \cdot \frac{\partial}{\partial r} \left(\frac{1}{r} \right) - \frac{\partial u}{\partial r} \cdot \frac{1}{r} \right] dS. \quad (103)$$

We want to show that the right side of the above equation tends to $4\pi u(\mathbf{0})$ as $\epsilon \rightarrow 0$. To do this, first notice that

$$\frac{\partial}{\partial r} \left(\frac{1}{r} \right) = -\frac{1}{r^2} = -\frac{1}{\epsilon^2}. \quad (104)$$

So the right side equals

$$\frac{1}{\epsilon^2} \iint_{r=\epsilon} u dS + \frac{1}{\epsilon} \iint_{r=\epsilon} \frac{\partial u}{\partial r} dS = 4\pi \bar{u} + 4\pi \epsilon \frac{\partial u}{\partial r}, \quad (105)$$

where the bar above represents the average value. Then, letting $\epsilon \rightarrow 0$, the expression approaches

$$4\pi u(\mathbf{0}) + 4\pi \cdot 0 \cdot \frac{\partial u}{\partial r}(\mathbf{0}) = 4\pi u(\mathbf{0}) \quad (106)$$

and we are done. $\square$